

MEESHO - JEWELRY MARKETPLACE DATABASE

CHAPTER – 1

Overview

The **Meesho Jewelry Marketplace Database** is designed to manage all the essential operations of a multi-vendor online jewelry marketplace. The database stores information related to customers, sellers, jewelry products, product categories, shopping carts, orders, payments, shipments, and product reviews. Each entity represents a real-world object in the marketplace and is connected to other entities through well-defined relationships. The use of Primary Keys and Foreign Keys ensures data integrity and minimizes data redundancy.

1. Customer Entity

Description

The **Customer** entity stores information about users who purchase jewelry products from the marketplace. Every customer has a unique identity and can browse products, add items to the cart, place orders, make payments, and submit product reviews.

Attributes

Attribute	Description
Customer_ID	Unique identification number of the customer.
Customer_Name	Name of the customer.
Email	Customer's email address.
Phone_Number	Contact number of the customer.
Password	Login password.
Address	Delivery address of the customer.

Primary Key Customer_ID

Relationships

- One customer can place many orders.
- One customer can add many products to the cart.
- One customer can write multiple product reviews.

2. Seller Entity

Description

The **Seller** entity stores information about jewelry sellers who list and sell products on the marketplace.

Attributes

Attribute	Description
Seller_ID	Unique identification number of the seller.
Seller_Name	Seller's full name.
Store_Name	Name of the jewelry store.
Email	Seller's email address.
Phone_Number	Contact number.
Address	Business address.

Primary Key Seller_ID

Relationships

- One seller can sell multiple jewelry products.
- One seller manages product inventory.

3. Category Entity

Description

The **Category** entity organizes jewelry products into different groups, making it easier for customers to browse products.

Attributes

Attribute	Description
Category_ID	Unique category identifier.
Category_Name	Name of the category (Necklace, Earrings, Rings, Bangles, etc.).

Primary Key Category_ID

Relationships

- One category contains many products.
- Each product belongs to one category.

4. Product Entity

Description

The **Product** entity stores complete details of every jewelry item available in the marketplace.

Attributes

Attribute	Description
Product_ID	Unique product identifier.
Product_Name	Name of the jewelry product.
Description	Product description.
Price	Selling price.
Stock	Available quantity.
Image	Product image path or URL.
Seller_ID	Seller who listed the product.
Category_ID	Product category.

Primary Key Product_ID

Foreign Keys

- Seller_ID
- Category_ID

Relationships

- A seller can list many products.
- A category contains many products.
- A product can appear in many customer carts.
- A product can receive multiple reviews.

5. Cart Entity

Description

The **Cart** entity stores products selected by customers before placing an order.

Attributes

Attribute	Description
Cart_ID	Unique cart identifier.
Customer_ID	Customer who owns the cart.
Product_ID	Selected product.
Quantity	Number of items added.

Primary Key Cart_ID

Foreign Keys

- Customer_ID
- Product_ID

Relationships

- One customer has one shopping cart.
- One cart contains multiple products.

6. Orders Entity

Description

The **Orders** entity stores all order transactions placed by customers.

Attributes

Attribute	Description
Order_ID	Unique order identifier.
Customer_ID	Customer who placed the order.
Order_Date	Date of purchase.
Total_Amount	Total order amount.

Order_Status	Pending, Packed, Shipped, Delivered, etc.
--------------	---

Primary Key Order_ID

Foreign Key

- Customer_ID

Relationships

- One customer can place multiple orders.
- Each order belongs to one customer.
- Each order has one payment record.
- Each order has one shipment record.

7. Payment Entity

Description

The **Payment** entity stores payment details for every customer order.

Attributes

Attribute	Description
Payment_ID	Unique payment identifier.
Order_ID	Associated order.
Payment_Method	UPI, Card, Net Banking, Wallet, COD.
Payment_Status	Paid, Pending, Failed.
Payment_Date	Date of payment.

Primary Key Payment_ID

Foreign Key

- Order_ID

Relationships

- Each payment belongs to one order.
- Every completed order has one payment record.

8. Shipment Entity

Description

The **Shipment** entity stores delivery information for customer orders.

Attributes

Attribute	Description
Shipment_ID	Unique shipment identifier.
Order_ID	Associated order.
Delivery_Address	Customer delivery address.
Shipment_Status	Packed, Shipped, Out for Delivery, Delivered.
Delivery_Date	Date of delivery.

Primary Key Shipment_ID

Foreign Key

- Order_ID

Relationships

- Each shipment belongs to one order.
- Every order has one shipment record.

9. Review Entity

Description

The **Review** entity stores customer feedback and ratings for jewelry products.

Attributes

Attribute	Description
Review_ID	Unique review identifier.
Customer_ID	Customer who submitted the review.
Product_ID	Reviewed product.

Rating	Rating (1–5 stars).
--------	---------------------

Review_Text Customer comments.

Primary Key Review_ID

Foreign Keys

- Customer_ID
- Product_ID

Relationships

- One customer can write many reviews.
- One product can receive many reviews.

Entity Relationships Summary

Entity	Relationship	Related Entity
Customer	Places	Orders
Customer	Owns	Cart
Customer	Writes	Review
Seller	Sells	Product
Category	Contains	Product
Product	Belongs To	Category
Cart	Contains	Product
Orders	Has	Payment
Orders	Has	Shipment
Product	Receives	Review

Meesho Jewelry Marketplace Database

Entity	Attributes	Primary Key	Foreign Key	Relationship
Customer	Customer_ID, Customer_Name, Email, Phone_Number, Password, Address	Customer_ID	—	Owns Cart, Places Order, Writes Review, Makes Payment
Seller	Seller_ID, Seller_Name, Store_Name, Email, Phone_Number, Address	Seller_ID	—	Lists Product, Manages Inventory, Receives Order
Category	Category_ID, Category_Name, Category_Description	Category_ID	—	Categorizes Product
Product	Product_ID, Product_Name, Description, Price, Stock, Image, Seller_ID, Category_ID	Product_ID	Seller_ID, Category_ID	Belongs to Category, Listed by Seller, Appears in Order_Item, Receives Review, Added to Cart
Cart	Cart_ID, Customer_ID, Product_ID, Quantity	Cart_ID	Customer_ID, Product_ID	Owned by Customer, Contains Product
Order	Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status	Order_ID	Customer_ID	Placed by Customer, Includes Order_Item, Has Payment, Has Shipment
Order_Item	Order_Item_ID, Order_ID, Product_ID, Quantity, Price	Order_Item_ID	Order_ID, Product_ID	Belongs to Order, Refers to Product
Payment	Payment_ID, Order_ID, Payment_Method, Payment_Status, Payment_Date	Payment_ID	Order_ID	Belongs to Order

Entity	Attributes	Primary Key	Foreign Key	Relationship
Shipment	Shipment_ID, Order_ID, Delivery_Address, Shipment_Status, Delivery_Date	Shipment_ID	Order_ID	Belongs to Order, Delivered to Customer
Review	Review_ID, Customer_ID, Product_ID, Rating, Review_Text, Review_Date	Review_ID	Customer_ID, Product_ID	Written by Customer, Given for Product
Inventory	Inventory_ID, Seller_ID, Product_ID, Stock_Quantity, Last_Updated	Inventory_ID	Seller_ID, Product_ID	Managed by Seller, Stores Product
Administrator	Admin_ID, Admin_Name, Email, Phone_Number	Admin_ID	—	Manages Customers, Sellers, Products, Categories, and Orders